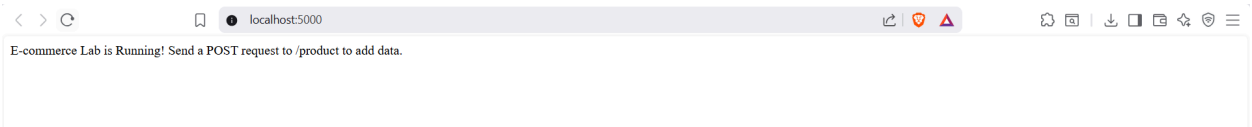
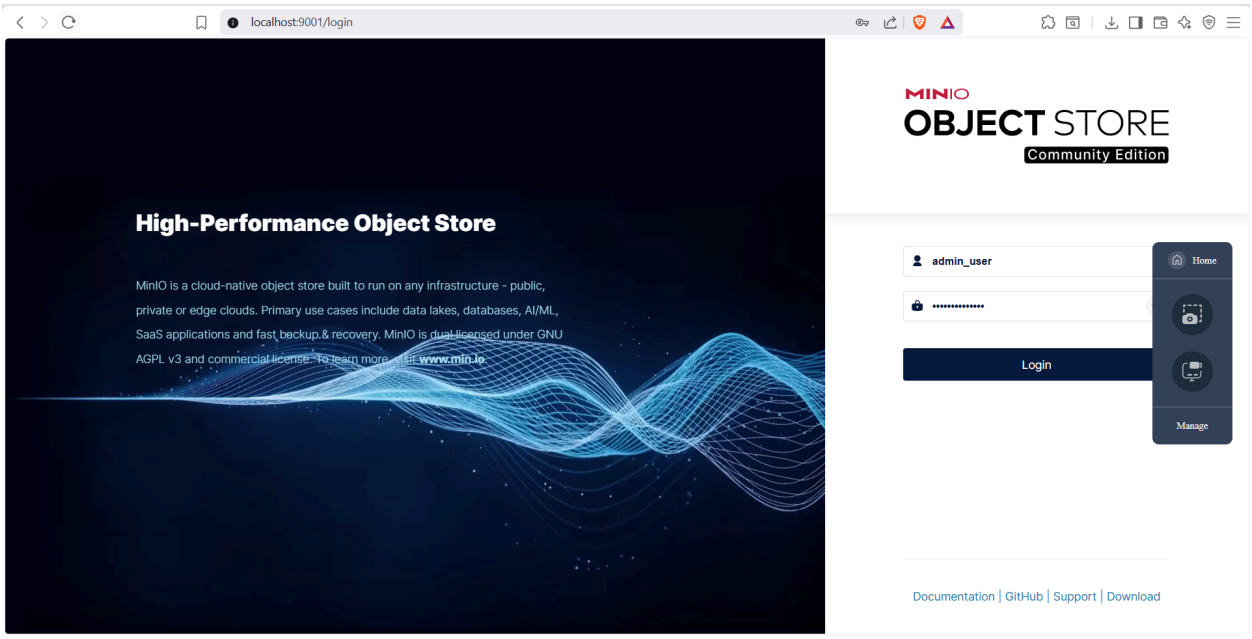
CC
Lab 7

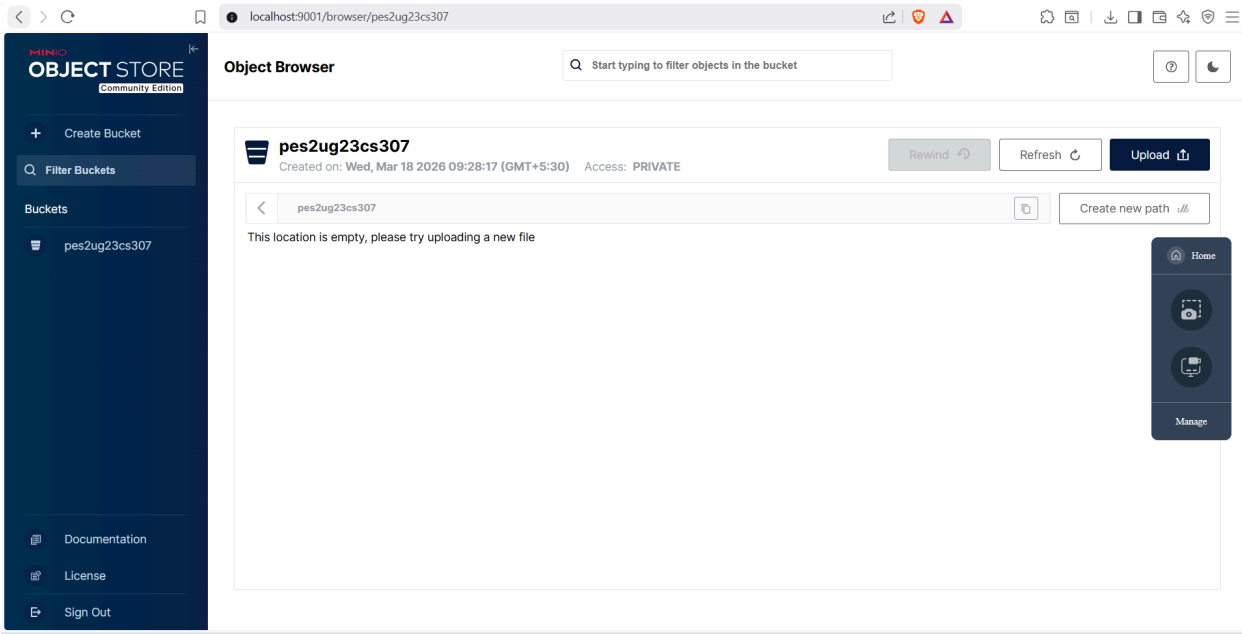
SS1



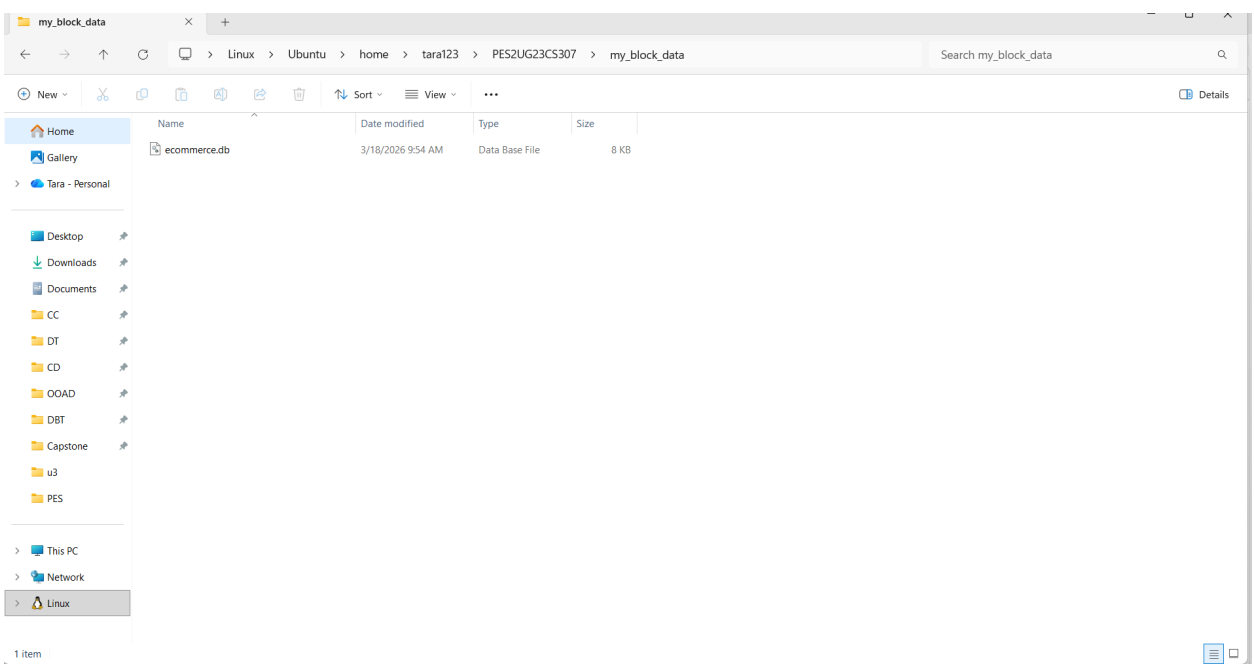
SS2



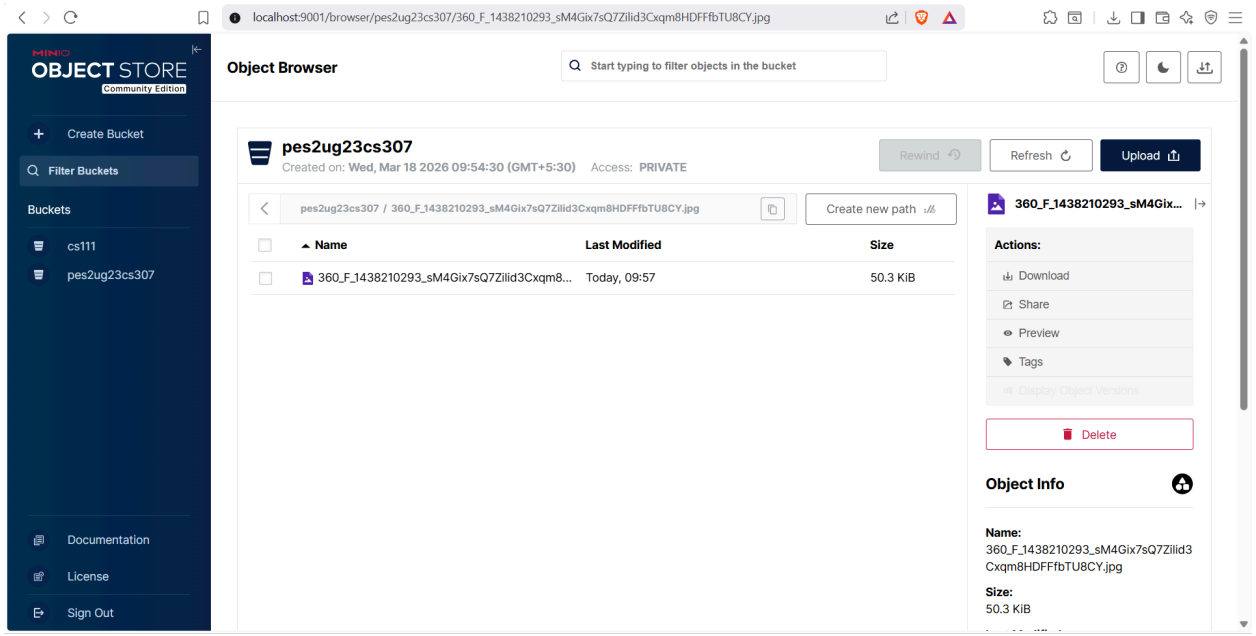
SS3




```
tara123@Tara:~/PES$ curl -X POST \
  -F "name=Phone" \
  -F "price=500" \
  -F "image=@360_F_1438210293_sM4Gix7sQ7Zilid3Cxqm8HDFFfbTU8CY.jpg" \
  http://localhost:5000/product
{"msg":"Structured data in Block, Image in Object","status":"success"}
```



```
tara123@Tara:~/PES2UG23CS307$ docker exec -it pes2ug23cs307-webapp-1 python -c "
import sqlite3;
conn = sqlite3.connect('/mnt/block_volume/ecommerce.db');
cursor = conn.cursor();
cursor.execute('SELECT * FROM product');
print(cursor.fetchall());
conn.close()
"
[(1, 'Phone', 500.0, '360_F_1438210293_sM4Gix7sQ7Zilid3Cxqm8HDFFfbTU8CY.jpg'), (2, 'Phone', 500.0, '360_F_1438210293_sM4
Gix7sQ7Zilid3Cxqm8HDFFfbTU8CY.jpg'), (3, 'Phone', 500.0, '360_F_1438210293_sM4Gix7sQ7Zilid3Cxqm8HDFFfbTU8CY.jpg'), (4, '
Phone', 500.0, '360_F_1438210293_sM4Gix7sQ7Zilid3Cxqm8HDFFfbTU8CY.jpg'), (5, 'Phone', 500.0, '360_F_1438210293_sM4Gix7sQ
7Zilid3Cxqm8HDFFfbTU8CY.jpg'), (6, 'Phone', 500.0, '360_F_1438210293_sM4Gix7sQ7Zilid3Cxqm8HDFFfbTU8CY.jpg')]
tara123@Tara:~/PES2UG23CS307$
```



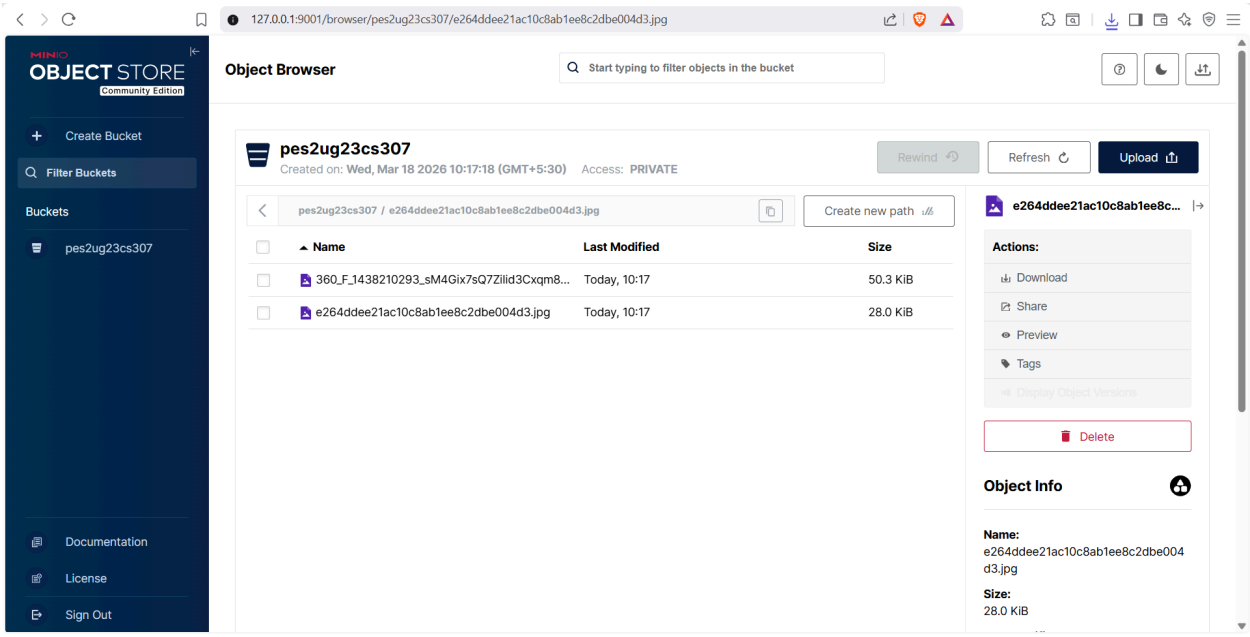
Experiment 2 :

```python
try:
    bucket = "pes2ug23cs307"
    if not minio_client.bucket_exists(bucket):
        minio_client.make_bucket(bucket)

    file_metadata = {
        "x-amz-meta-product-name": str(name),
        "x-amz-meta-product-price": str(price)
    }

    minio_client.fput_object(
        bucket,
        image.filename,
        temp_path,
        metadata=file_metadata
    )

    new_product = Product(name=name, price=price, image_name=image.filename)
    db.session.add(new_product)
    db.session.commit()
```



Experiment 3 :

```
tara123@Tara:~/PES2UG23CS307/my_block_data$ sudo fallocate -l 50M dummy1.bin
sudo fallocate -l 50M dummy2.bin
[sudo] password for tara123:
```

Name	Date modified	Type	Size
dummy1.bin	3/18/2026 10:23 AM	BIN File	51,200 KB
dummy2.bin	3/18/2026 10:23 AM	BIN File	51,200 KB
ecommerce.db	3/18/2026 10:13 AM	Data Base File	8 KB

Block storage relies on a fixed disk volume. As the storage fills up, the system may eventually encounter errors such as *"database full"* or disk write failures when attempting to insert new records into the database. This demonstrates that block storage is limited by physical storage capacity.

Experiment 4 :

```
tara123@Tara:~/PES2UG23CS307$ docker-compose stop minio
WARN[0000] /home/tara123/PES2UG23CS307/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, plea
se remove it to avoid potential confusion
[+] stop 1/1
 ✔Container pes2ug23cs307-minio-1 Stopped                                                                          0.7s
tara123@Tara:~/PES2UG23CS307$ curl -X POST \
  -F "name=Phone" \
  -F "price=500" \
  -F "image=@yourimage.jpg" \
  http://localhost:5000/product
curl: (26) Failed to open/read local data from file/application
tara123@Tara:~/PES2UG23CS307$
```

```
tara123@Tara:~/PES2UG23CS307$ docker-compose start minio
WARN[0000] /home/tara123/PES2UG23CS307/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, plea
se remove it to avoid potential confusion
[+] start 1/1
 ✔Container pes2ug23cs307-minio-1 Started                                                                          0.8s
tara123@Tara:~/PES2UG23CS307$ docker-compose stop webapp
WARN[0000] /home/tara123/PES2UG23CS307/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, plea
se remove it to avoid potential confusion
[+] stop 1/1
 ✔Container pes2ug23cs307-webapp-1 Stopped                                                                         1.5s
tara123@Tara:~/PES2UG23CS307$
```

```
tara123@Tara:~/PES2UG23CS307$ cd my_block_data
tara123@Tara:~/PES2UG23CS307/my_block_data$ ls -lh
total 101M
-rw-r--r-- 1 root root  50M Mar 18 04:53 dummy1.bin
-rw-r--r-- 1 root root  50M Mar 18 04:53 dummy2.bin
-rw-r--r-- 1 root root 8.0K Mar 18 04:43 ecommerce.db
tara123@Tara:~/PES2UG23CS307/my_block_data$
```